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Rigid body mechanics

Problem Set VII

Exercise 1:

A homogeneous solid sphere S , of mass m , radius a , and center G , rolls without slipping on a fixed horizontal plane Π , located at elevation $z = 0$, relative to a frame $R(O, X, Y, Z)$, with basis $(\vec{i}, \vec{j}, \vec{k})$, assumed to be Galilean. The OZ axis is vertical upward. We denote by H the intersection point of plane Π with the OZ axis. We denote by I the geometric contact point of S with Π and $\vec{\Omega}$ the instantaneous rotation vector of S with respect to R . The torsor of contact actions at point I is reduced to the reaction $\vec{R}$.

1. Write the rolling without slipping relation of S on plane Π , then deduce the velocity and acceleration vectors of the center of mass G , with respect to R , as a function of $\vec{\Omega}$ and $\dot{\vec{\Omega}}$.
2. Determine the time-dependent vector equation of motion of G , $\overrightarrow{HG}(t)$, by applying the general theorems. What is the nature of the motion of G ? We denote by $\vec{v}_0$ the initial velocity vector of G and by $\vec{r}_0$ the initial position vector of G .

In what follows, we assume that plane Π is rotating about the OZ axis and we denote by $\vec{\omega}$ the instantaneous rotation vector of the plane with respect to R .

3. Determine the velocity of center of mass G , $\vec{v}_G$, using the rolling without slipping relation.
4. Find, using the general theorems, the acceleration of center of mass G , $\vec{\gamma}_G$, as a function of $\vec{\omega}$ and $\vec{\Omega}$ and their time derivatives. Deduce a first vector integral of motion.

Let r and φ be the polar coordinates of G , defined in the plane XOY by $\overrightarrow{HG} = r\vec{u}_r$. Initially, we have the following values $r_0 = r(0)$, $\dot{r}_0 = \dot{r}(0)$, and $\dot{\varphi}_0 = \dot{\varphi}(0)$. And we denote by μ the sliding friction coefficient of S on plane Π .

5. Determine the time-dependent equations of motion $r(t)$ and $\varphi(t)$, and deduce $\overrightarrow{HG}(t)$ as a function of time.
6. Determine $\vec{R}$ and $\vec{M}_I$ and indicate the condition that μ must satisfy for rolling without slipping to persist.

Correction**1. Rolling without slipping relation and velocity/acceleration of G:**

The rolling without slipping condition means the velocity of the contact point I relative to the plane is zero:

$$\vec{V}_I(S/R) = \vec{0}$$

Using the velocity distribution formula:

$$\vec{V}_I(S/R) = \vec{V}_G(S/R) + \vec{\Omega} \times \vec{GI}$$

Since $\vec{GI} = -a\vec{k}$ (assuming the sphere is above the plane), we have:

$$\vec{V}_G(S/R) + \vec{\Omega} \times (-a\vec{k}) = \vec{0}$$

$$\vec{V}_G(S/R) = a\vec{\Omega} \times \vec{k}$$

This is the rolling without slipping relation.

Now, differentiating with respect to time to get acceleration:

$$\vec{\gamma}_G(S/R) = \frac{d\vec{V}_G}{dt} = a \frac{d\vec{\Omega}}{dt} \times \vec{k} + a\vec{\Omega} \times \frac{d\vec{k}}{dt}$$

Since $\vec{k}$ is fixed in the Galilean frame R , $\frac{d\vec{k}}{dt} = \vec{0}$, so:

$$\vec{\gamma}_G(S/R) = a\dot{\vec{\Omega}} \times \vec{k}$$

2. Equation of motion of G and nature of motion:

Apply the momentum theorem:

$$m\vec{\gamma}_G = m\vec{g} + \vec{R}$$

Where $\vec{R}$ is the reaction at the contact point.

The angular momentum theorem about point G :

$$\frac{d\vec{\sigma}_G}{dt} = \vec{M}_G(\vec{R})$$

For a sphere: $\vec{\sigma}_G = I_G\vec{\Omega} = \frac{2}{5}ma^2\vec{\Omega}$

And $\vec{M}_G(\vec{R}) = \vec{GI} \times \vec{R} = (-a\vec{k}) \times \vec{R}$

So:

$$\frac{2}{5}ma^2\dot{\vec{\Omega}} = -a\vec{k} \times \vec{R}$$

$$\dot{\vec{\Omega}} = -\frac{5}{2ma}\vec{k} \times \vec{R}$$

Now substitute into the acceleration expression:

$$\vec{\gamma}_G = a \left(-\frac{5}{2ma}\vec{k} \times \vec{R} \right) \times \vec{k} = -\frac{5}{2m}(\vec{k} \times \vec{R}) \times \vec{k}$$

Using the vector triple product identity: $(\vec{A} \times \vec{B}) \times \vec{C} = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{B} \cdot \vec{C})\vec{A}$

So: $(\vec{k} \times \vec{R}) \times \vec{k} = (\vec{k} \cdot \vec{k})\vec{R} - (\vec{R} \cdot \vec{k})\vec{k} = \vec{R} - R_z\vec{k}$

Therefore:

$$\vec{\gamma}_G = -\frac{5}{2m}(\vec{R} - R_z \vec{k})$$

From the momentum theorem: $m\vec{\gamma}_G = m\vec{g} + \vec{R}$

Substitute:

$$\begin{aligned} m \left[-\frac{5}{2m}(\vec{R} - R_z \vec{k}) \right] &= m\vec{g} + \vec{R} \\ -\frac{5}{2}(\vec{R} - R_z \vec{k}) &= m\vec{g} + \vec{R} \\ -\frac{5}{2}\vec{R} + \frac{5}{2}R_z \vec{k} &= m\vec{g} + \vec{R} \\ -\frac{7}{2}\vec{R} + \frac{5}{2}R_z \vec{k} &= m\vec{g} \end{aligned}$$

Separate into horizontal and vertical components:

Vertical: $-\frac{7}{2}R_z + \frac{5}{2}R_z = -mg \Rightarrow -R_z = -mg \Rightarrow R_z = mg$

Horizontal: $-\frac{7}{2}\vec{R}_h = \vec{0} \Rightarrow \vec{R}_h = \vec{0}$

So the reaction is purely vertical: $\vec{R} = mg\vec{k}$

Then from the momentum theorem: $m\vec{\gamma}_G = m\vec{g} + mg\vec{k} = \vec{0}$

So $\vec{\gamma}_G = \vec{0}$, meaning the center of mass moves with constant velocity.

The equation of motion is: $\overrightarrow{HG}(t) = \overrightarrow{HG}(0) + \vec{v}_0 t$

The motion of G is uniform rectilinear motion.

3. Velocity of G with rotating plane:

When the plane rotates with angular velocity $\vec{\omega} = \omega\vec{k}$, the rolling without slipping condition becomes that the velocity of the contact point relative to the rotating plane is zero.

Let R' be the frame rotating with the plane. The condition is:

$$\vec{V}_I(S/R') = \vec{0}$$

Using the composition of velocities:

$$\vec{V}_I(S/R) = \vec{V}_I(S/R') + \vec{V}_I(R'/R)$$

But $\vec{V}_I(R'/R) = \vec{\omega} \times \overrightarrow{HI}$

And $\overrightarrow{HI} = \overrightarrow{HG} + \overrightarrow{GI} = \vec{r} - a\vec{k}$, where $\vec{r} = \overrightarrow{HG}$

So: $\vec{V}_I(R'/R) = \vec{\omega} \times (\vec{r} - a\vec{k}) = \omega\vec{k} \times \vec{r}$ (since $\vec{k} \times \vec{k} = \vec{0}$)

The rolling condition $\vec{V}_I(S/R') = \vec{0}$ gives:

$$\vec{V}_I(S/R) = \omega\vec{k} \times \vec{r}$$

But also: $\vec{V}_I(S/R) = \vec{V}_G(S/R) + \vec{\Omega} \times \overrightarrow{GI} = \vec{V}_G + \vec{\Omega} \times (-a\vec{k})$

So: $\vec{V}_G + \vec{\Omega} \times (-a\vec{k}) = \omega\vec{k} \times \vec{r}$

Therefore: $\vec{V}_G = a\vec{\Omega} \times \vec{k} + \omega\vec{k} \times \vec{r}$

4. Acceleration of G and first integral:

Differentiate the velocity expression:

$$\vec{\gamma}_G = \frac{d\vec{V}_G}{dt} = a\dot{\vec{\Omega}} \times \vec{k} + a\vec{\Omega} \times \frac{d\vec{k}}{dt} + \dot{\omega}\vec{k} \times \vec{r} + \omega\vec{k} \times \vec{V}_G$$

Since $\vec{k}$ is fixed and assuming constant ω :

$$\vec{\gamma}_G = a\dot{\vec{\Omega}} \times \vec{k} + \omega\vec{k} \times \vec{V}_G$$

Now apply the momentum theorem:

$$m\vec{\gamma}_G = m\vec{g} + \vec{R}$$

And the angular momentum theorem about G :

$$\frac{d\vec{\sigma}_G}{dt} = \vec{M}_G(\vec{R}) = \vec{G}I \times \vec{R} = (-a\vec{k}) \times \vec{R}$$

With $\vec{\sigma}_G = I_G\vec{\Omega} = \frac{2}{5}ma^2\vec{\Omega}$, so:

$$\frac{2}{5}ma^2\dot{\vec{\Omega}} = -a\vec{k} \times \vec{R}$$

$$\dot{\vec{\Omega}} = -\frac{5}{2ma}\vec{k} \times \vec{R}$$

Substitute into the acceleration:

$$\begin{aligned}\vec{\gamma}_G &= a \left(-\frac{5}{2ma}\vec{k} \times \vec{R} \right) \times \vec{k} + \omega\vec{k} \times \vec{V}_G \\ &= -\frac{5}{2m}(\vec{k} \times \vec{R}) \times \vec{k} + \omega\vec{k} \times \vec{V}_G\end{aligned}$$

Using the vector identity again: $(\vec{k} \times \vec{R}) \times \vec{k} = \vec{R} - R_z\vec{k}$

So:

$$\vec{\gamma}_G = -\frac{5}{2m}(\vec{R} - R_z\vec{k}) + \omega\vec{k} \times \vec{V}_G$$

From momentum theorem: $m\vec{\gamma}_G = m\vec{g} + \vec{R}$

So:

$$\begin{aligned}m \left[-\frac{5}{2m}(\vec{R} - R_z\vec{k}) + \omega\vec{k} \times \vec{V}_G \right] &= m\vec{g} + \vec{R} \\ -\frac{5}{2}(\vec{R} - R_z\vec{k}) + m\omega\vec{k} \times \vec{V}_G &= m\vec{g} + \vec{R} \\ -\frac{7}{2}\vec{R} + \frac{5}{2}R_z\vec{k} + m\omega\vec{k} \times \vec{V}_G &= m\vec{g}\end{aligned}$$

Separate components:

Vertical: $-\frac{7}{2}R_z + \frac{5}{2}R_z = -mg \Rightarrow -R_z = -mg \Rightarrow R_z = mg$

Horizontal: $-\frac{7}{2}\vec{R}_h + m\omega\vec{k} \times \vec{V}_G = \vec{0}$

So: $\vec{R}_h = \frac{2}{7}m\omega\vec{k} \times \vec{V}_G$

Now the momentum theorem gives:

$$m\vec{\gamma}_G = m\vec{g} + mg\vec{k} + \frac{2}{7}m\omega\vec{k} \times \vec{V}_G$$

$$\vec{\gamma}_G = \frac{2}{7}\omega\vec{k} \times \vec{V}_G$$

This is the equation of motion. To find a first integral, take the dot product with $\vec{V}_G$:

$$\vec{\gamma}_G \cdot \vec{V}_G = \frac{2}{7}\omega(\vec{k} \times \vec{V}_G) \cdot \vec{V}_G = 0$$

Since $(\vec{k} \times \vec{V}_G) \cdot \vec{V}_G = 0$ (perpendicular vectors).

So $\frac{d}{dt}(\frac{1}{2}V_G^2) = 0$, meaning the speed is constant: $V_G = \text{constant}$

This is the first integral.

5. Equations of motion in polar coordinates:

In polar coordinates: $\vec{r} = r\vec{u}_r$, $\vec{V}_G = \dot{r}\vec{u}_r + r\dot{\varphi}\vec{u}_\varphi$

The acceleration is:

$$\vec{\gamma}_G = (\ddot{r} - r\dot{\varphi}^2)\vec{u}_r + (r\ddot{\varphi} + 2\dot{r}\dot{\varphi})\vec{u}_\varphi$$

From the equation of motion: $\vec{\gamma}_G = \frac{2}{7}\omega\vec{k} \times \vec{V}_G$

Now $\vec{k} \times \vec{V}_G = \vec{k} \times (\dot{r}\vec{u}_r + r\dot{\varphi}\vec{u}_\varphi) = \dot{r}\vec{u}_\varphi - r\dot{\varphi}\vec{u}_r$

So:

$$\vec{\gamma}_G = \frac{2}{7}\omega(\dot{r}\vec{u}_\varphi - r\dot{\varphi}\vec{u}_r) = -\frac{2}{7}\omega r\dot{\varphi}\vec{u}_r + \frac{2}{7}\omega\dot{r}\vec{u}_\varphi$$

Equate components:

Radial: $\ddot{r} - r\dot{\varphi}^2 = -\frac{2}{7}\omega r\dot{\varphi}$

Angular: $r\ddot{\varphi} + 2\dot{r}\dot{\varphi} = \frac{2}{7}\omega\dot{r}$

These are the equations of motion.

From the first integral, we know $V_G^2 = \dot{r}^2 + r^2\dot{\varphi}^2 = \text{constant}$

The position vector is: $\vec{HG}(t) = r(t)\vec{u}_r(\varphi(t))$

6. Reaction $\vec{R}$ and moment $\vec{M}_I$, condition on μ :

We already found:

$$\vec{R} = \vec{R}_h + R_z\vec{k} = \frac{2}{7}m\omega\vec{k} \times \vec{V}_G + mg\vec{k}$$

For rolling without slipping to persist, the friction force must satisfy the Coulomb condition:

$$\|\vec{R}_h\| \leq \mu R_z = \mu mg$$

So:

$$\left\| \frac{2}{7}m\omega\vec{k} \times \vec{V}_G \right\| \leq \mu mg$$

$$\frac{2}{7}\omega V_G \leq \mu g$$

$$\mu \geq \frac{2\omega V_G}{7g}$$

This is the condition on μ .

The moment at point I is zero for a perfect rolling without slipping contact (no friction moment for a sphere on a plane).

Therefore, the torsor of contact actions at I is:

$$\tau(I) = [\vec{R}, \vec{0}]$$

with $\vec{R} = \frac{2}{7}m\omega\vec{k} \times \vec{V}_G + mg\vec{k}$